

Population and Sampling

The target population consists of graduates of PLP from academic years 2023, 2024, and 2025. The study utilizes a stratified purposive sampling approach, where respondents are grouped by cohort year to ensure representation. A minimum response rate of 50% per cohort is targeted to ensure statistical validity, with higher targets depending on cohort size:

- Small cohorts (<100): 70%
- Medium cohorts (100–300): 60%
- Large cohorts (>300): 50%
- **Step 1: Compute Mean Scores per Domain**
- Survey responses are measured using a 5-point Likert scale in Table 2.
 - **Table 2.** Five-point Likert Scale for Survey Responses

Response	Score
Strongly Agree	5
Agree	4
Neutral	3
Disagree	2
Strongly Disagree	1

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- The score for each domain is computed as the mean of the indicator items.
$$D_j = \frac{\sum_{i=1}^n X_{ij}}{n}$$
- **Formula (Equation 1):**
- Where:
 - D_j = Domain mean score for domain j
 - X_{ij} = Individual Item Score of indicator i in domain j
 - n = Number of indicators in the domain
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- D_j – This is the average score of a specific domain
- Example domains: Employability (E), Family (F)
- So, D_E = domain mean of Employability and D_F = domain mean of Family
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- X_{ij} – This represents the Individual item response (score of respondent i on item j)
- It means Each response in the questionnaire (Likert scale)
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- X_{ij} is the *Sum of Scores*. Just Add all item scores in that domain: $3+2+3+2+4 = 14$
- n – This is the number of items in the domain

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- **Example:** Employability (E) Items
- Compute D_j^{Before} and D_j^{After}

Item	Score Before	Score After
Q1	3	5
Q2	2	4
Q3	3	4
Q4	2	3
Q5	4	4

- Putting it together,
- $D_j^{\text{Before}} = \frac{\sum_{i=1}^n X_{ij}}{n} = \frac{3+2+3+2+4}{5} = 2.80$
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- $D_j^{\text{After}} = \frac{\sum_{i=1}^n X_{ij}}{n} = \frac{5+4+4+3+4}{5} = 4.00$
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- The average employability score **before** graduation is 2.80. The average employability score **after** graduation is 4.00.

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- **Step 2: Compute GAIN Score**
- **Formula (Equation 2):**
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$$G_j = D_j^{\text{After}} - D_j^{\text{Before}}$$

- Where:
- G_j = Gain score of domain j
- D_j^{After} = domain mean AFTER score
- D_j^{Before} = domain mean BEFORE score
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- This shows how much improvement happened due to education.
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- Example: Employability (E),
 - $G_E = 4.0 - 2.80 = 1.20$
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- **Step 3: Compute the Weighted Gain** $w_j \times G_j$
- Where:
- G_j = Gain score of domain j
- w_j = weight assigned to domain j
- G_j – This is the **improvement** in each domain. It shows how much the student improved

- w_j – This represents the **importance of each domain**.
- $w_j G_j$ – This tells you how much this domain contributes to overall impact
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- Example:
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Domain	Gain	Weight	Weighted $w_j G_j$
Employability	1.20	0.30	0.36
Family	1.00	0.25	0.25
Personal	1.10	0.20	0.22
Civic	0.70	0.15	0.105
Governance	0.80	0.10	0.08

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- **Step 4: Compute the Overall PEII Score**

$$PEII = \sum_{j=1}^5 w_j G_j$$

- **Formula (Equation 3):**
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- or expanded version:
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- **Formula (Equation 4):**
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$$PEII = (0.30G_E) + (0.25G_F) + (0.20G_P) + (0.15G_C) + (0.10G_G)$$

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- Substituting the Weighted $w_j G_j$ above
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- - $PEII = 0.36 + 0.25 + 0.22 + 0.105 + 0.08 = \mathbf{1.015}$
- The PEII is a weighted sum of gain scores across domains, reflecting the overall impact of education on graduates' lives. For example, a PEII of 1.015 indicates moderate improvement across areas such as employability, personal development, and family support.
- **Step 5: Compute Cohort → Base-100 Index**
- To enable meaningful comparison of educational impact across different cohorts, the computed mean PEII scores are transformed into a **Base-100 Index**. This standardizes all cohort results relative to a selected reference group (baseline cohort).

Formula (Equation 5):

$$PEII\ Index_c = \left(\frac{\overline{PEII}_c}{\overline{PEII}_{base}} \right) \times 100$$

Where:

PEII IndexC = Index value of cohort c

PEIIc = Mean PEII score of cohort c

PEIIbase = Mean PEII score of the *base cohort* (e.g., 2023)

Since cohorts consist of different individuals, the index reflects differences in average improvement, not causal effects or superiority. External factors (e.g., economic conditions, labor market changes) may also influence cohort outcomes. The PEII Index is used for institutional monitoring, not for measuring individual improvement

Example:

Cohort	Mean PEII	Index	Interpretation
2023(Base)	1.015	100	Reference level
2024	1.715	118.23	18.23% higher average improvement
2025	0.90	88.67	11.33% lower average improvement

The base cohort (2023) has a mean PEII of 1.015 and is assigned an index value of 100 for standardization. Subsequent cohorts are compared relative to this baseline using proportional transformation. The following are computational breakdown:

For **2023 (Base)**: $1.015 \times 100 = 100$

For **2024**: $1.715 \times 100 = 171.5$

For **2025**: $0.90 \times 100 = 90$